

Classifying Sentiments on Social Media Texts: A GPT-4 Preliminary Study

Lany L. MACEDA*

Bicol University, Legazpi City, Philippines

Miles B. Artiaga

Bicol University, Legazpi City, Philippines

Jennifer L. Llovido

Bicol University, Legazpi City, Philippines

Mideth B. Abisado

National University, Manila, Philippines

ABSTRACT

In today's digital age, social media has become a hub for people to express their thoughts and feelings. Sentiment classification discerns public opinions and trends to understand their sentiments towards a certain topic. Often, achieving accurate sentiment classifications in large datasets necessitate the use of human-annotated training data which can be costly and time-consuming. Large Language Models (LLMs) like the Generative Pre-trained models by OpenAI have surged in popularity due to its capabilities in understanding the given tasks. In this preliminary study, we report the performance of the latest OpenAI GPT-4 using zero- and one-shot learning approaches on classifying sentiments when fed with social media dataset. Notably, the latter approach written in English which mimics the instructions designed for human annotators, achieved a substantial agreement ($k = 0.77$) with human annotations, displaying high accuracy, precision, and recall accordingly even without explicit training data. Meanwhile, the fine-tuned mBERT resulted to lower evaluation scores than the GPT-4. Our findings provide foundational insights into the strengths and limitations of GPT-4 for sentiment classification in a social media dataset, setting the groundwork for broad future research in this field.

CCS CONCEPTS

• Computing methodologies; • Artificial Intelligence; • Natural Language Processing; • Natural Language Generation;

KEYWORDS

GPT-4, Sentiment Annotation, LLM Prompting, Social Media Data

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1 INTRODUCTION

Social media platforms have become virtual hubs where people share their ideas, sentiments, and opinions on a wide range of topics

[4, 19], creating a large corpus of unlabeled and unstructured data. Such data are often unstructured, multilingual, multicultural [1], and involves the use of slangs, acronyms, or emoticons. In contrast to newswire and weblogs, there is almost no subsequent editing or filtering of the incoming textual content [30]. These occurrences become bottlenecks for many Natural Language Processing (NLP) tasks including classifying of sentiments [17].

Sentiment analysis is a subfield of NLP that automatically evaluates natural language expressions, identifies key assertions or viewpoints, and categorizes them based on their emotional stance [34]. The insights produced can be instrumental in making informed decisions in real life situations [17]. For instance, [18] and [2] recommended continuous improvement of educational programs in the Philippines. These studies undergo rigorous manual annotation of each document performed by field experts to establish a gold standard set to train a machine or deep learning model.

In 2017, the introduction of Transformers [31], has since become the foundation of various Large Language Models (LLMs) [3, 6] in NLP [15]. These models utilize computational techniques and statistical approaches to produce text resembling human language, allowing for a deep understanding of context and the expression of dynamic concepts [25]. LLMs are tasked to perform tasks using carefully crafted natural language prompts. According to [37], this paradigm could be valuable because it may enable generation of high-quality annotations with less demand for human annotators due to its state-of-the-art performance on various NLP tasks.

The Generative Pre-trained Transformer 4 (GPT-4), launched in March 2023 by OpenAI [22], is a Transformer-based model that has undergone pre-training to predict the subsequent token in a document. This training utilizes a combination of publicly accessible data (such as internet content) and data licensed from third-party providers. Although OpenAI have claimed state-of-the-art results of GPT-4 on difficult based on the conducted preliminary model evaluation, they reported limited information regarding the model's architecture (including its size), the hardware used, the computational resources for training, the dataset creation process, the training methodology, or related details, due to the competitive environment and the safety concerns associated with extensive models like GPT-4 [22].

In this preliminary study, we aim to contribute foundation for understanding the potential application of an advanced language model in the field of sentiment analysis, by evaluating GPT-4 on social media texts and compare its performance to human-annotated dataset. Specifically, we utilize the human-annotated social media data related to the implementation of the Philippine Universal Access to Quality and Tertiary Education (UAQTE). We emphasize that

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this dataset encompasses the use of slangs, emoticons, and multiple low-resourced languages within a single sentence such as Filipino, Bicol, and other native languages in the Philippines, a phenomenon known as code-mixing. Considering the nature of this dataset, we further compare the evaluation results to Multilingual Bidirectional Encoder Representations from Transformers (mBERT) [6] model and perform manual error analysis to show the capabilities and limitations of GPT-4 for sentiment analysis task.

2 RELATED WORKS

2.1 Sentiment Classification on Social Media Texts

Social media platforms provide vast amount of user-generated content that can be used to gauge public opinion or sentiments on various topics. However, low-resource languages have received less attention due to the scarcity of annotated datasets, limited access to computational resources, and shortage of experts in the field [36] including the Philippines and its native languages. There are two most commonly used approaches for sentiment classification: supervised learning and unsupervised learning [14, 20]. In the supervised learning approach, a machine learning or deep learning algorithm is usually trained to classify sentiments using a manually annotated dataset in which human annotators have identified the sentiment of each document. For instance, [18] used mBERT to classify large code-mixed texts from various social media platforms such as Twitter, Facebook, and YouTube, achieving the highest accuracy of 80.21% using a human-annotated training set. Similarly, [10] used mBERT to classify the sentiments of social media posts which achieved the F1-scores of 0.603, 0.698, and 0.595 for the Tamil, Malayalam, and Kannada code-mixed languages, respectively. In the study of [29], text data underwent comprehensive preprocessing steps, including emoji, repeated characters, and punctuation removal. Traditional machine learning, deep learning, transfer learning, and hybrid deep learning models were then compared before and after these processes. The CNN+BiLSTM hybrid deep learning model achieved the highest accuracy of 0.66 on preprocessed Tamil code-mixed data.

Alternatively, unsupervised techniques may incur minimal resource costs, but the outcomes they yield can often lack clarity and interpretability [14], as exemplified by the utilization of lexicon-based approaches. These existing lexicons are composed of mainly words that are deemed to carry sentiments but can be quite limited when applied to new tasks due to numerous out-of-vocabulary words [30]. According to [35], instead of training task-specific models, a pre-trained LLM could be used directly for various tasks without fine-tuning through prompt learning.

In this study, we compare the evaluation results of GPT-4 against the fine-tuned Multilingual BERT (mBERT), a multilingual version of BERT trained on 104 languages including English and widely used local languages in the Philippines such as Tagalog, Waray, and Cebuano.

2.2 GPT Models

Several studies have assessed the capabilities of LLM like GPT in various tasks, [7] conducted various annotation task and showed that

ChatGPT, a chatbot that uses GPT model, outperformed Mechanical Turk human annotators. Similarly, [9] reported that ChatGPT demonstrated great potential as a data annotation tool even with simple prompt design. In cost-effective scenarios, GPT-3 labeling can outperform human labeling, and incorporating a combination of GPT-3 and human labeling can lead to further enhancements in performance [32]. In [13], emphasized the effectiveness of GPT-3.5 in overcoming the standard linguistic nuances of sentiment analysis, such as contextualization and sarcasm detection. We also observed trends in assessing GPT-4 in various licensing examinations [11], [12] and medical problems [11, 15, 21].

These studies have focused on using monolingual datasets or publicly available data. It remains uncertain whether these open datasets are included in the model’s training data or not. In their technical report, OpenAI acknowledged the possible risk of unintended contamination on their benchmark assessments, where the model might have inadvertently gain access to the test questions and their corresponding answers [22]. Further, in the findings of [5], LLMs like GPT-4 and GPT-3.5 can undergo significant transformations within a relatively brief period, emphasizing the importance of ongoing monitoring of the model for possible enhancements. In utilizing a GPT-4 model to classify social media texts, specifically Tweets related to the topic of abortion legalization in the United States of America, [16] found that the Few-shot approach method exhibits a higher degree of similarity to human experts.

To the best of our knowledge, there have been no studies that have conducted sentiment classification on a social media dataset using the GPT-4 model, particularly on datasets that are both low-resourced and code-mixed.

3 METHODOLOGY

3.1 Data

Due to our limited access to GPT-4, we opted to work with a dataset comprising 200 unprocessed data points for each sentiment category: neutral, negative, and positive, sourced from [18]. This selection resulted in a total of 600 samples. These data points were gathered from Facebook, Twitter, and YouTube, specifically in the context of UAQTE. Each data sample was manually annotated by human experts, with a sentiment label denoted by values [0, 1, 2], which correspond to neutral, negative, and positive sentiments, respectively, and prepended a number for traceability of results.

As discussed in Section 2, it is worth noting that social media texts differ from conventional attribute-value data due to their inherently noisy, decentralized, unstructured, and dynamically evolving nature [8].

3.2 Prompt Template Design

We first experimented two prompt variations written in English. For this, we employ zero and n -shot learning approaches, where n specifies the number of examples used in a prompt.

For our zero-shot learning approach, we designed a straightforward prompt which follows a prompt structure of [Instruction/Constraints/ n samples]: “Classify the sentiments of the following texts. For each item, respond only 0, 1, or 2 to indicate whether the sentiment is neutral, negative, or positive, respectively. In cases of ambiguity, choose the prevailing sentiment”. For the second prompt,

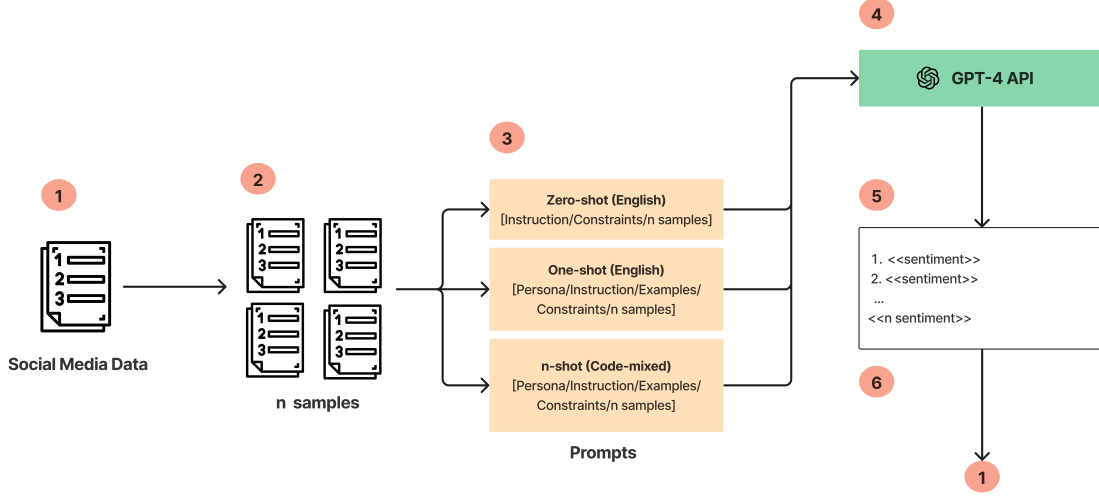


Figure 1: Sentiment Annotation Process using GPT-4 API. (1) Each data sample is prepended with a number; (2) Get n chunk of data samples; (3) Include the n data samples in the prompt; (4) Access the GPT-4 model through calling its API using the parameters set; (5) Results are printed and saved in a .csv file; and (5) Loop through the same process until n samples

we mimic the annotation instructions designed for the human annotators as our prompt instruction and provided a brief sentiment description and one example for each sentiment category, hence referred to as one-shot learning. This approach follows a similar prompt structure to zero-shot, except that we added identity modifier or persona “*You are the CHED-UniFAST program coordinator in the Philippines*” as our System’s prompt which was used to set the context of the dialogue, followed by the instruction and examples. The final prompt structure of one-shot learning is [Persona/Instruction/Examples/Constraints/ n samples].

In [33] suggested to evaluate different prompt designs such as use of code-mixed prompt. We selected the successful learning approach as the basis for our code-mixed prompt. To write the code-mixed prompt, we engaged with a native Bicolano speaker to translate the English prompt using their own comprehension and natural language, free from restrictions regarding usage of symbols or abbreviations. The final code-mixed prompt consisted the use of various languages, slang, abbreviation, and misspellings: “*Iclassify mo ung sintemyento netong mga social media texts gamit an mga minasanod na polarities about sa pag implement ng Philippine RA 10931 UAQTE Act program. . .*”. This approach allowed us to capture the informal and dynamic language style prevalent in our dataset.

3.3 Sentiment Annotation using GPT-4

The original GPT model [23] is composed of 12-layer decoder-only Transformer with masked self-attention heads. Its primary capability is fine-tuning for specific downstream tasks.

As of the writing, GPT-4 is by far the most advanced model released by OpenAI. GPT-4’s text input capability was released via ChatGPT and the API. In this study, we use the GPT-4 API with 8,192 tokens context model, comprising of both prompt and completion which entails corresponding price per token. During our

initial test runs, GPT-4 is costly especially with longer prompts. To reduce our costs, we employed looping of batch processing, which means that for each request, it will process 30 and 15 samples from the dataset for zero- and one-shot prompts respectively, instead of individually sending each data sample to API. This API allows us to communicate with the model by specifying a set of parameters that control its behavior during text generation, such as a temperature of 0, and a 200 maximum output tokens for every API call. We only run the experiment once as our final evaluation and save the output in a .csv file. Figure 1 summarizes our sentiment annotation process using GPT-4 API.

3.4 Evaluation

Following the evaluation of [28] to analyze the agreement or inter-rater reliability among annotators, we used Cohen’s Kappa. Table 1 shows the interpretation of the Kappa value obtained. Additionally, we employed commonly used classification metrics such as accuracy, precision, and recall to evaluate the model’s predictive accuracy and its ability to correctly classify sentiments. Accuracy measures the overall correctness of model predictions, while precision focuses on the proportion of true positives among positive predictions, and recall quantifies the ability of the model to capture all positive instances.

Further, we compare the evaluation results against the fine-tuned Multilingual BERT (mBERT). It is worth noting that the dataset used for both models underwent no preprocessing. For fine-tuning mBERT a 70:30 split ratio for training and testing sets, a learning-rate and batch size of $2e-5$ and 16 were employed. Both mBERT and GPT are Transformer based pre-trained models.

Table 1: Interpretation of Cohen’s Kappa

Kappa value	Interpretation
<0.20	Poor agreement
0.21-0.40	Fair agreement
0.41-0.60	Moderate agreement
0.61-0.80	Substantial agreement
0.81-1.00	Very good agreement

4 RESULTS AND DISCUSSION

In Table 2, we present the performance of GPT-4 using various learning approaches in a classification task, each evaluated based on Cohen’s Kappa value, accuracy, precision, and recall. We run the experiment once as our final evaluation due to our limited access to GPT-4 API and increased cost associated with longer prompts.

Compared to the achieved score of the prompt that uses zero-shot approach, the Kappa values of prompts written in English and Code-mixed using the one-shot approach achieved 0.77 and 0.73 respectively, which indicate that there is a substantial and statistically significant level of agreement between the model’s predictions and the human annotations if context and identity modifier are given. Accordingly, these prompts achieved an accuracy score of 0.85 and 0.8250 which demonstrates a high proportion of correct classifications, emphasizing the capability of the model’s predictions even without explicit training data to learn the sentiments of an unprocessed code-mixed dataset. The matching precision and recall values signify that the model maintains a balanced approach, effectively capturing positive instances while minimizing false positives. Meanwhile, the fine-tuned mBERT resulted to lower results than the GPT-4. This result is similar to the study of [24], where the GPT-3 model beats the BERT-base model in Marathi Text Classification.

As stated by [16], GPT models are highly sensitive to prompts. Interestingly, the code-mixed prompt achieved scores that were not significantly different from the prompt written in English. We initially conclude that the use of words or language in the prompt may not be the primary issue, as long as the prompt provides identity modifier (System’s prompt), adequate instructions, and context.

4.1 Error Analysis

We collected five samples from our annotated dataset to manually examine instances where the model’s predictions diverged from or did not match the human-annotated data. Our primary observation was despite our explicit instruction to respond in numerical

values [0, 1, 2] which corresponds to sentiments: neutral, negative, and positive, the one-shot prompts received responses in words, sometimes numerical values, unlike the zero-shot prompt which responded the correct format for all items. Hence, post-cleaning of the collected responses was necessary to evaluate GPT-4’s performance on various metrics. These inaccuracies and inconsistencies have been also observed in the study of [27].

In Table 3 we focused on the results of GPT-4 model; the first sample written in non-English text was annotated correctly by all prompts; the second and third example, utilizing a one-shot prompt, correctly identified the text implying an announcement even with the inclusion of positive phrase like “*kayo ang bidai!*”. This pattern was consistently observed in other samples, emphasizing the significance of incorporating constraints for improved classification accuracy. While one-shot prompt achieved high evaluation scores, it still predicted inaccurate sentiments as shown in the fourth and fifth examples where the latter uses a Filipino slang word “*cutie*” which means “*to manifest something they want to have/achieve*” [26]. We hypothesize that this could be due to potential influence of the model’s pre-trained knowledge that might introduce interpretative variations, particularly in a domain-specific contexts. Interestingly, mBERT correctly classified all data samples despite its low evaluation scores.

5 CONCLUSION

In this preliminary study, we explored the application of GPT-4 for classifying sentiments in social media texts. The results demonstrated GPT-4’s capability in handling diverse linguistic inputs without using any training data, provided that the prompt should contain specific and adequate instructions. However, challenges were identified in instances involving slang or domain-specific language.

6 LIMITATIONS AND FUTURE WORKS

Despite the impressive evaluation score obtained by GPT-4 in annotating the dataset, inaccuracies were observed on our experiments. We initially suggest employing GPT-4 to participate in the annotation process, alongside human annotators. The GPT-4’s performance is dependent to the dataset they were trained on and to the quality of the annotated data used for evaluation. Any biases from these datasets may directly affect the model’s performance. Additionally, the model’s effectiveness may vary across different domains and industries, as it might not be well-suited to highly specialized subjects or contexts not adequately covered in this experiment.

In this study, code-mixed dataset was used to evaluate the GPT-4 model capabilities in sentiment classification, since significant

Table 2: Performance of GPT-4 and mBERT model

Learning Approach	Kappa value	Accuracy	Precision	Recall
GPT-Zero-shot (English)	0.50	0.6683	0.6870	0.6683
GPT-One-shot (English)	0.77	0.85	0.8665	0.85
GPT-One-shot (Code-mixed)	0.73	0.8250	0.8432	0.8250
mBERT	-	0.8277	0.8381	0.8252

Table 3: Examples of Classified Sentiments by GPT-4 and mBERT

No.	Human	mBERT	Zero-shot	One-shot (EN)	One-shot (CM)	Examples
1	Negative	Negative	Negative	Negative	Negative	Kasuya man kan bank Portal! Subago pa Password expired dai na nag untok.hahaha. (EN: The bank portal is frustrating! My password has been expired for a while now, and it's not getting any better.hahaha)
2	Neutral	Neutral	Positive	Neutral	Positive	TES Update #85 October pa lang parang Pasko na sa UniFAST! TES Cuties ng SOCCSKSARGEN, kayo ang bida ngayong alas otso gabi! (EN: TES Update #85, it's only October, but it already feels like Christmas at UniFAST! TES Cuties of SOCCSKSARGEN, you're the stars tonight at eight o'clock!)
3	Neutral	Neutral	Positive	Neutral	Neutral	Gawis ay agsapa, PINASKolars! Reminding again our TES Beneficiaries to answer our #quickpoll and Google Form that you can locate by clicking this #tesgrant. Iyaman!
4	Negative	Negative	Negative	Neutral	Negative	Hello po just wanna ask po kung kailan po kaya ang realese ng TDP? Medyo matagal na din po kasi yunh sakin di ko pa rin po natatanggap Salamat po (EN: Hello, I just want to ask when will the TDP be released? It's been a while, and I still haven't received it. Thank you.)
5	Positive	Positive	Positive	Neutral	Neutral	Ched Unifast Cutie

results were obtained from the experiments, this suggests that future studies could focus on testing GPT-4 on purely-written native and less-common languages, promoting inclusivity and broadening the scope of sentiment analysis.

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